

CSES — Geometry Problem Reference

All tasks: 1 s, 512 MiB. Coordinates and answers are exact integers unless stated otherwise.

1740 — Intersection Points

Given n horizontal/vertical segments, count their intersection points. Parallel segments never intersect, and no segment endpoint is an intersection point. **In:** n ; then endpoints $(x_1, y_1), (x_2, y_2)$

Out: number of intersections

Constraints: $n \leq 10^5$; coordinates in $[-10^6, 10^6]$; each segment is nonempty.

1741 — Area of Rectangles

Given n axis-aligned rectangles, find the total area of their union; overlapping area is counted once. **In:** n ; then opposite corners $(x_1, y_1), (x_2, y_2)$

Out: union area

Constraints: $n \leq 10^5$; $-10^6 \leq x_1 < x_2 \leq 10^6$, same for y .

1742 — Robot Path

A robot starts at $(0, 0)$ and follows axis-aligned commands. It stops after all commands or immediately upon reaching any point visited earlier, possibly partway through a command. Find distance traveled until stopping. **In:** n ; then direction $d \in \{U, D, L, R\}$ and distance x

Out: total distance traveled

Constraints: $n \leq 10^5$, $1 \leq x \leq 10^6$.

2189 — Point Location Test

For directed line $p_1 \rightarrow p_2$, classify point p_3 as lying to its left, right, or on the line. **In:** t ; then p_1, p_2, p_3 coordinates

Out: LEFT, RIGHT, or TOUCH per test

Constraints: $t \leq 10^5$; coordinates in $[-10^9, 10^9]$; $p_1 \neq p_2$.

2190 — Line Segment Intersection

For two closed nondegenerate segments, determine whether they share at least one point. Endpoint touching and collinear overlap both count as intersection. **In:** t ; then endpoints of both segments

Out: YES/NO per test

Constraints: $t \leq 10^5$; coordinates in $[-10^9, 10^9]$.

2191 — Polygon Area

Given the vertices of a simple polygon in boundary order, compute its area exactly. **In:** n ; then vertices (x_i, y_i)

Out: $2A$, where A is the polygon area

Constraints: $3 \leq n \leq 1000$; coordinates in $[-10^9, 10^9]$.

2192 — Point in Polygon

Given a simple polygon and m query points, classify each point as inside, outside, or on the polygon boundary. **In:** n, m ; n polygon vertices; then m points

Out: INSIDE, OUTSIDE, or BOUNDARY per point

Constraints: $n, m \leq 1000$; coordinates in $[-10^9, 10^9]$.

2193 — Polygon Lattice Points

For a simple polygon with integer-coordinate vertices, count lattice points strictly inside it and lattice points on its boundary.

In: n ; then integer vertices (x_i, y_i)

Out: two integers: interior count, boundary count

Constraints: $3 \leq n \leq 10^5$; coordinates in $[-10^9, 10^9]$.

2194 — Minimum Euclidean Distance

Among n distinct planar points, find the minimum Euclidean distance between two different points. **In:** n ; then points (x, y)

Out: d^2 for the minimum distance d

Constraints: $2 \leq n \leq 2 \cdot 10^5$; coordinates in $[-10^9, 10^9]$.

2195 — Convex Hull

Find the convex hull of n distinct points. Output *all* input points lying on the hull boundary, including collinear points along hull edges; any output order is accepted. **In:** n ; then points (x, y)

Out: k , then the k hull-boundary points

Constraints: $3 \leq n \leq 2 \cdot 10^5$; hull area is positive; coordinates in $[-10^9, 10^9]$.

3410 — Maximum Manhattan Distances

Points are inserted one by one into an initially empty set. After each insertion, report the maximum Manhattan distance $|x_1 - x_2| + |y_1 - y_2|$ between any two inserted points. **In:** n ; then the points in insertion order

Out: one maximum distance after each insertion

Constraints: $n \leq 2 \cdot 10^5$; points distinct; coordinates in $[-10^9, 10^9]$.

3411 — All Manhattan Distances

Given n distinct points, compute the sum of Manhattan distances over all unordered pairs. **In:** n ; then points (x, y)

Out: $\sum_{i < j} (|x_i - x_j| + |y_i - y_j|)$

Constraints: $n \leq 2 \cdot 10^5$; coordinates in $[-10^9, 10^9]$.

3427 — Line Segments Trace I

Each segment runs from $(0, y_1)$ to (m, y_2) and has integer slope. For every integer $x = 0, 1, \dots, m$, find the maximum y attained by any segment there. **In:** n, m ; then y_1, y_2 for each segment

Out: $m + 1$ maxima for $x = 0, \dots, m$

Constraints: $n, m \leq 10^5$, $0 \leq y_1, y_2 \leq 10^9$.

3428 — Line Segments Trace II

Segments have integer-coordinate endpoints, integer slope, and arbitrary active intervals $[x_1, x_2] \subseteq [0, m]$. For every integer x , find the maximum y among segments covering that x ; output -1 if none does. **In:** n, m ; then $(x_1, y_1), (x_2, y_2)$

Out: $m + 1$ values for $x = 0, \dots, m$

Constraints: $n, m \leq 10^5$; $0 \leq x_1 < x_2 \leq m$; $0 \leq y_1, y_2 \leq 10^9$.

3429 — Lines and Queries I

Process online queries: 1 **a b** adds the infinite line $y = ax + b$; 2 **x** asks the maximum value among all lines at that x . The first query is always an insertion. **In:** n ; then n queries

Out: maximum value for each type 2 query

Constraints: $n \leq 2 \cdot 10^5$; $|a|, |b| \leq 10^9$; $0 \leq x \leq 10^5$.

3430 — Lines and Queries II

Process online queries: 1 **a b l r** adds $y = ax + b$ active only for $x \in [l, r]$; 2 **x** asks the maximum among lines active at x . **In:** n ; then n queries

Out: maximum for each type 2 query, or NO if no line is active

Constraints: $n \leq 2 \cdot 10^5$; $|a|, |b| \leq 10^9$; $0 \leq x, l, r \leq 10^5$, $l \leq r$.